DETERMINANTS ON ONE PAGE

Why i makes Fibonacci inevitable

A determinant subtracts, and the imaginary unit turns its subtraction into the addition that defines Fibonacci.

1 / THE RULE AND THE PICTURE

The tridiagonal continuant subtracts x^2, and $x = i$ makes it add

$$\begin{bmatrix} 1 & x & 0 & 0 \\ x & 1 & x & 0 \\ 0 & x & 1 & x \\ 0 & 0 & x & 1 \end{bmatrix}$$

$K_4(x)$

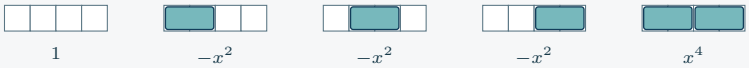
Expanding along the first row gives the continuant recurrence, whose minus sign comes from the alternating character of the determinant, and setting $x = i$ gives $-x^2 = -i^2 = +1$:

$$K_n = K_{n-1} - x^2 K_{n-2} \xrightarrow{x=i} K_n = K_{n-1} + K_{n-2}$$

n	1	2	3	4	5	6	7
$K_n(i)$	1	2	3	5	8	13	21

2 / WHY THE SUM IS A COUNT / SQUARES AND DOMINOES

Each term of the determinant is a tiling of a strip, and every domino weighs $-x^2$



$K_4 = 1 - 3x^2 + x^4$, and at $x = i$ the five tilings each weigh $+1$, so $K_4(i) = 1 + 3 + 1 = 5$.

In general $K_n(x) = \sum_k \binom{n-k}{k}(-x^2)^k$, one term per tiling with k dominoes, so $K_n(i) = \sum_k \binom{n-k}{k} = F_{n+1}$, the shallow diagonals of Pascal's triangle. The planar version of this trick is Kasteleyn's: weighting the vertical edges by i makes a determinant count domino tilings of a rectangle, the model behind Catalan's constant.

3 / THE METALLIC FAMILY / TWO WAYS TO CHANGE THE MATRIX

The amplitude $\sqrt{m}\,i$ weighs dominoes, and the diagonal d weighs squares

matrix	recurrence	first terms	growth
diagonal 1, off-diagonal i	$D_n = D_{n-1} + D_{n-2}$	$1, 2, 3, 5, 8, 13$	$\varphi = 1.618$
diagonal 1, off-diagonal $\sqrt{2}\,i$	$D_n = D_{n-1} + 2D_{n-2}$	$1, 3, 5, 11, 21, 43$ (Jacobsthal)	2
diagonal 1, off-diagonal $\sqrt{3}\,i$	$D_n = D_{n-1} + 3D_{n-2}$	$1, 4, 7, 19, 40, 97$	2.303
diagonal 2, off-diagonal i	$D_n = 2D_{n-1} + D_{n-2}$	$2, 5, 12, 29, 70$ (Pell)	$1 + \sqrt{2} = 2.414$
diagonal 3, off-diagonal i	$D_n = 3D_{n-1} + D_{n-2}$	$3, 10, 33, 109$	$\frac{3+\sqrt{13}}{2} = 3.303$

The recurrence $D_n = dD_{n-1} + D_{n-2}$ has growth $\lambda^2 = d\lambda + 1$, the d-th metallic mean (golden, silver, bronze), so Pell and the silver mean come from the diagonal 2 and not from the amplitude $\sqrt{2}\,i$. Tridiagonal matrices only link neighbours and so give second order; Tribonacci needs a 3×3 companion matrix.

4 / CASSINI / THE DYNAMIC DETERMINANT

Consecutive Fibonacci numbers span a parallelogram of area one

With $Q = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ we have $Q^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$ and $\det Q = -1$, so

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n, \qquad \text{for example } 8 \cdot 3 - 5^2 = -1, \quad 13 \cdot 5 - 8^2 = +1$$

The columns stretch as the numbers grow, yet the signed area stays at ± 1 and its sign records a flip of handedness at every step.

5 / THE LADDER OF NORMS / MATRIX SYMMETRY SELECTS THE NUMBER SYSTEM

Change the structure of the matrix, and the determinant lands in a different ring

matrix	determinant	ring	integers produced
$\begin{pmatrix} a & b \\ b & a \end{pmatrix}$	$a^2 - b^2$	split	odd and multiples of 4
$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$	$a^2 + b^2$	$\mathbb{Z}[i]$	sums of two squares
3×3 circulant	$a^3 + b^3 + c^3 - 3abc = (a+b+c)\,N$	$\mathbb{Z}[\omega]$	not $\equiv 3, 6 \pmod 9$

Here $N = a^2 + b^2 + c^2 - ab - bc - ca = p^2 - pq + q^2$ with $p = a - c$, $q = b - c$ is the Eisenstein norm; the triple $(n+1, n, n-1)$ gives $9n$, and if $3 \mid \det$ then $3 \mid s = a + b + c$ and $q \equiv s^2 \equiv 0$, so $9 \mid \det$.

ASIDE / WHAT i DOES

It does not create the recurrence; it selects the additive branch of a structure that was already there.

ASIDE / CHECK IN PYTHON

`det(np.eye(4)+1j*(np.eye(4,k=1)+np.eye(4,k=-1)))` gives 5.

Convention. $K_0 = K_1 = 1$ and $F_1 = F_2 = 1$, so $K_n(i) = F_{n+1}$. After *Why i Makes Fibonacci Inevitable* (McMurmerings).